

# Wout Goesaert

Astronomy PhD Candidate at Leiden University



## About me

I am a PhD student at Leiden University. My current research focuses on studying the evolved phases of stellar evolution in quiescent galaxies using NIR observations.

My passion for astronomy began early, prompting me to join a youth astronomy organisation in Belgium. This organisation ultimately inspired my academic career.

In 2019, I moved to The Netherlands to study astronomy and physics in Leiden, where I have found great joy in collaborating with peers and working alongside inspiring scientists.

I enjoy giving talks, participating in scientific discussions, and stargazing with my Dobsonian telescope.

## Areas of specialization

Radio • Infrared • AGNs  
• Molecular Astronomy  
• ALMA • JWST

 personal website

 Wout Goesaert

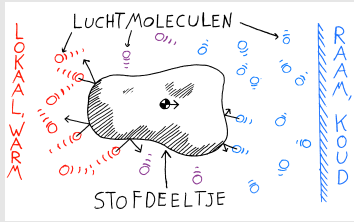
 woutgoesaert

## RESEARCH PROJECTS

- |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 2025—2029 | <b>TP-AGB stars as actors of galaxy evolution</b><br><b>PHD</b> · Leiden University <br>Supervisor: Prof. Mariska K.T. Kriek                                                                                                                                                                                                                                                                                                                                             |    |
| 2024      | <b>ALMA: Resolving the torus and outflow of the Circinus AGN</b><br><b>RESEARCH INTERNSHIP</b> · ESO: Santiago de Chile <br>Supervisors: Dr. Konrad R. W. Tristram and Dr. C.M. Violette Impellizzeri<br>Working on a publication that builds on my first master's thesis with a focus on the outflows and inflows traced by the continuum and molecular kinematics. Attempting to construct the first mass balance equation between a SMBH, torus, CND and galaxy.    |    |
| 2023—2024 | <b>JWST: H<sub>3</sub><sup>+</sup> Absorption and Emission Seen by JWST Probes Extremely High Ionisation Rates in (U)LIRGs: 10/10</b><br><b>MASTER THESIS</b> · Leiden University <br>Supervisors: Prof.dr. P.P. van der Werf and V.A. Buiten MSc.<br>Utilised JWST data to study H <sub>3</sub> <sup>+</sup> in (U)LIRGs, leading to the <b>discovery of H<sub>3</sub><sup>+</sup> emission in the ISM</b> and the highest measured ionisation rates to date.         |    |
| 2022—2023 | <b>ALMA: A novel view on the Circinus torus and circumnuclear disk: 9.5/10</b><br><b>FIRST MASTER PROJECT</b> · Leiden University <br>Supervisors: Dr. C.M. Violette Impellizzeri and Dr. Alex P.S. Hygate<br>Cleaned, self-calibrated and combined ALMA radio data. First measurements of the (column)density and temperature in a torus (LTE + non-LTE). Studied line ratios, continuum mechanism, morphology and kinematics at highest spatial resolution to date. |   |
| 2022      | <b>MeerKAT: Polarimetry of Galaxy Cluster Magnetic Fields: 9/10</b><br><b>BACHELOR THESIS</b> · Leiden University <br>Supervisors: Dr. Reinout J. van Weeren and Dr. Erik Osinga.<br>Mapping Faraday rotation of radio sources in Abell 85 using the MeerKAT Galaxy Cluster Legacy Survey. Determined the feasibility and limits of using the MGCLS catalog for this purpose.                                                                                        |  |
| 2021—2022 | <b>Quantum Optics: Delayed Choice Quantum Eraser: 9.5/10</b><br><b>EXPERIMENTAL PROJECT</b> · Leiden University <br>Supervisors: Dr. Wolfgang Löffler and Matteo Fiscaro MSc<br>Built a delayed choice quantum eraser by storing superposition in polarisation of entangled photons in a team of seven. Demonstrated Bell inequality for the first time in Leiden. Took on both a scientific and managing role in the project.                                       |  |
| 2020—2023 | <b>Fluid dynamics: Capillary waves on water jets</b><br><b>FIRST AUTHOR PUBLICATION</b> · Leiden University <br>Co-author: Dr. Ir. Paul Logman<br>Published in the American Journal of Physics. Developed a new method to accurately measure the dispersion relation of capillary waves on water jets and constructed a semi-quantitative simulation for educational purpose.                                                                                        |  |

## PUBLICATIONS

- |                 |                                                                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2025</b>     | Goesaert, W. M. et al. Torus feeding and outflow launching in the active nucleus of the Circinus galaxy. <a href="https://doi.org/10.1051/0004-6361/202555556">https://doi.org/10.1051/0004-6361/202555556</a>                                                  |
| <b>2024</b>     | Goesaert, W. M., & Logman, P. S. W. M. (2024). Easy method to establish the dispersion relation of capillary waves on water jets. <i>American Journal of Physics</i> , 92(2). <a href="https://doi.org/10.1119/5.0144849">https://doi.org/10.1119/5.0144849</a> |
| <b>in prep.</b> | Goesaert, W. M. et al. (in prep.). Strength of TP-AGB Emission in Distant Quiescent Galaxies.                                                                                                                                                                   |



**Popular scientific writing:** A wonderful world of physics lies right in front of your nose. Just look and ask "Why?" There is no need for a particle accelerator to do new physics! In my bimonthly column "Exceptioneel Ordinaire Fysica" for the Flemish journal "Astra", I show how extraordinary physics lurks behind ordinary everyday topics and phenomena. I made the illustration above for a recent article in this series on "thermophoresis", which explored the connection between dusty windowsills and an unsolved problem in planet formation.



I love combining my photography hobby with astronomy. This is a narrow-band image of the Crab Nebula that I made with the Isaac Newton Telescope during observing time that was unexpectedly left over.

## EDUCATION

|              |                                                                        |
|--------------|------------------------------------------------------------------------|
| 2025-present | <b>PhD in Astrophysics</b><br>· Leiden University                      |
| 2022-2025    | <b>Astronomy Research</b><br>Msc. · Leiden University<br>GPA: 9.0/10   |
| 2019-2022    | <b>Astronomy</b><br>Bsc. · Leiden University<br>Cum Laude, GPA: 8.8/10 |
| 2019-2022    | <b>Physics</b><br>Bsc. · Leiden University<br>Cum Laude, GPA: 8.8/10   |

## AWARDS & GRANTS

|             |                                                                                                |
|-------------|------------------------------------------------------------------------------------------------|
| <b>2025</b> | Janneke Fruin-Helb Subsidie — Leiden University Fund (LUF)                                     |
| <b>2024</b> | Scholarship for Chile research project by LUF                                                  |
| <b>2023</b> | 1st place Dutch physics Bsc. thesis contest — NNV-SPIN                                         |
| <b>2022</b> | Bronze Galilei Award — recognition for my commitment to the Belgian Astronomical Society (VVS) |
| <b>2019</b> | 1st place Flemish Astronomy Olympiad                                                           |
| <b>2019</b> | 5th place Flemish Physics Olympiad                                                             |
| <b>2015</b> | Young Scientist Award — UHasselt                                                               |

## EXPERIENCE, SERVICE & OUTREACH

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 2025–        | <b>Supervision</b><br>· Leiden, The Netherlands                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
|              | Supervision of Msc. students: Maria Kalitsounaki (Gas Outflows in Quiescent Galaxies at Cosmic Noon) and Yara Beekhuizen (The Nature of UV Radiation in Quiescent Galaxies).                                                                                                                                                                                                                                                                                                                           |  |
| 2020–2024    | <b>Observing experience</b><br>JWST, ISAAC NEWTON AND MERCATOR ·                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
|              | JWST observation planning for the NIRSPEC MOS micro shutter array. Service mode observations using HERMES at the Mercator Telescope (KU Leuven, 3 nights), remote observations with Isaac Newton Telescope (Leiden Uni, 2 nights).                                                                                                                                                                                                                                                                     |  |
| 2024         | <b>Teaching assistant</b><br>PLANETARY SYSTEMS · Leiden, The Netherlands                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
| 2024–present | <b>Organising committee Astronomy Olympiad</b><br>FLEMISH ASTRONOMY OLYMPIAD COMMITTEE · VVS, Belgium                                                                                                                                                                                                                                                                                                                                                                                                  |  |
| 2017–present | <b>Public Speaking</b><br>TALKS, LECTURE SERIES, CONFERENCES AND EVENTS<br>Audiences up to 300p, all age groups and backgrounds. Notable talks: 4 talks at ESO and UDP in Chile, focus talk at Nerdland Festival, yearly 10-part lecture series on astronomy at public observatory (2017-2023), VVS/JVS conferences (2020-2023), Astronomy on Tap Leiden, Cosmic Café Genk, public shows about my research in collaboration with musician NeoMaan, talk at ALMA + JWST Joint Science Day in Groningen. |  |
| 2016–present | <b>Popular science writing</b><br>SEVERAL COLUMNS AND ARTICLES IN POPULAR SCIENCE JOURNALS<br>Including: Exceptioneel Ordinaire Fysica (Astra magazine, 32 pieces), Wist-je-hopelijk-nog-niet-datjes (Astra magazine, 5 pieces) + several other publications at EOS, Selenia, Astra and Heelal.                                                                                                                                                                                                        |  |
| 2016–present | <b>President (5y) and board member (8y) at JVS-Descartes</b><br>BOARD MEMBER · Genk, Belgium<br>JVS-Descartes is an organisation with the aim of popularising physics and astronomy in children and youth. As president, I led an enthusiastic group of ~10 board members. My other functions included secretary, lecture series on astronomy, observation camp, observing, rocketry, PR, and organising several public events.                                                                        |  |
| 2018–2019    | <b>Guide at public observatory: Cosmodrome</b><br>TOUR GUIDE OBSERVATORY & PLANETARIUM · Genk, Belgium                                                                                                                                                                                                                                                                                                                                                                                                 |  |

## TECHNICAL SKILLS

I regularly use Python,  $\LaTeX$ , Linux, GitHub, JWST APT, CASA, CARTA, 3D-Barolo and DS9 for my research activities.

## LANGUAGES

|                |        |
|----------------|--------|
| <b>Dutch</b>   | native |
| <b>English</b> | C2     |
| <b>French</b>  | B1     |

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